

SAMVAAD-IQ

Conversational Crime Intelligence OS for Karnataka Police

Powered by KAVACH Crime DNA Engine and Multi-Agent Detective Room

Complete Prototype Build Document

Focused on Zoho Catalyst Deployment, Manual Test Data, Demo Flow, and Submission Requirements

Prepared for Hackathon Submission: Prototype Brief, GitHub Public Repository, Catalyst Deployed Link, Demo Video, Prototype Deck

Core Positioning

SAMVAAD-IQ is not a normal chatbot. It is a role-based investigation workspace where police users can ask English/Kannada/Kanglish questions, receive evidence-grounded answers, view maps and criminal networks, find similar cases through Crime DNA, simulate patrol deployment, and export legal-style PDF intelligence briefs. All demo records are synthetic and privacy-safe.

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1. Executive Summary

SAMVAAD-IQ is a complete prototype concept and implementation plan for Problem Statement 2: Intelligent Conversational AI for KSP Crime Database. The prototype converts synthetic FIR-style records into an interactive investigation environment. It supports natural-language crime queries, Kannada/Kanglish demo prompts, role-based login, dashboards, geospatial hotspot analysis, criminal relationship graphs, similar case matching, patrol what-if simulation, and evidence-backed PDF report export.

The aim is to submit a working prototype that can be deployed on Catalyst by Zoho, shared through a public Catalyst deployment link, documented in a public GitHub repository, and explained through a public demo video and prototype deck. The system is intentionally framed as an investigative decision-support tool, not as an automated guilt prediction system.

- Main brand: SAMVAAD-IQ.
- Core backend intelligence: KAVACH Crime DNA Engine.
- User interface style: Conversational Investigation OS with a visible Detective Room reasoning panel.
- Deployment focus: Catalyst Web Client Hosting + Catalyst Authentication + Catalyst Data Store + Functions/AppSail + optional SmartBrowz PDF.
- Dataset: Fully synthetic manual data to ensure the prototype responds reliably during evaluation.

2. Problem Statement Selection and Strategic Fit

2.1 Selected Challenge

Selected problem statement: Intelligent Conversational AI for KSP Crime Database.

2.2 Why PS2 is stronger than PS1

- PS1 is mainly analytics and visualization. Many teams can build maps, charts, and dashboards; the differentiation ceiling is lower.
- PS2 is harder because it combines RAG-style querying, multilingual interaction, graph reasoning, criminological insights, explainability, and operational decision support.
- PS2 naturally includes PS1 because maps, charts, dashboards, and relationship graphs become outputs of the conversational intelligence layer.
- A well-made PS2 prototype looks like a real investigation platform, not just a dashboard clone.

2.3 Prototype success definition

- A judge can log in as an Investigator or Supervisor.
- The dashboard loads meaningful synthetic crime records.
- The user can ask a crime question and receive a grounded answer from the data.
- The app can show a map, network graph, similar case matches, and a PDF-style report.
- The public link must be deployed using Catalyst by Zoho, as required by the submission form.

3. Hackathon Submission Mapping

Submission Field	What to Submit	SAMVAAD-IQ Preparation
Challenges	Select the problem statement from the hackathon portal.	Choose: Intelligent Conversational AI for KSP Crime Database.
Prototype Brief	Maximum 1024 characters.	Use the ready brief in Section 4.

GitHub Public Repository Link	Public repo URL starting with https/http.	Create repo with client, server, seed data, README, demo screenshots, and deployment notes.
Prototype Deployed Link	Catalyst platform by Zoho only.	Deploy React frontend and backend on Catalyst; paste the public Catalyst URL.
Demo Video Link Public	Google Drive public link or unlisted/private YouTube.	Record the 3-minute script in Section 16; make sharing public.
Upload Prototype Deck	PDF deck upload.	Use deck structure in Section 17 and export as PDF.

4. Prototype Brief for Submission Form

SAMVAAD-IQ is a conversational crime intelligence OS for Karnataka Police. It lets investigators ask English/Kannada/Kanglish questions over synthetic FIR-style records and instantly receive evidence-grounded answers, hotspot maps, criminal network graphs, similar case matches, patrol what-if insights, and PDF investigation briefs. The prototype uses role-based login, Catalyst Data Store seed data, a Crime DNA similarity engine, and a multi-agent Detective Room that separates retrieval, network analysis, skepticism, and reporting. It is designed as human-in-the-loop decision support, not automated guilt prediction, with audit logs, confidence notes, and legal-style source trails for accountable policing.

Character count: 713 / 1024

5. Product Vision and User Personas

5.1 One-line vision

SAMVAAD-IQ turns fragmented FIR records into a conversational, explainable, and visual crime intelligence workspace for proactive investigation.

5.2 User personas

Persona	Pain Point	SAMVAAD-IQ Value
Investigator	Searches manually across FIR files, Excel sheets, and station-level records.	Natural-language search, similar case discovery, network links, case report generation.
Crime Analyst	Needs trends, hotspots, repeat MO, and cross-district patterns.	Dashboard, map clusters, Crime DNA scoring, repeat offender patterns.
Supervisor / Commissioner	Needs strategic decision support and patrol planning.	What-if patrol simulation, district risk views, PDF briefs, audit logs.
Admin / SCRB Data Officer	Needs controlled access and clean data governance.	Role-based access, data seeding, audit logs, no real PII in prototype.

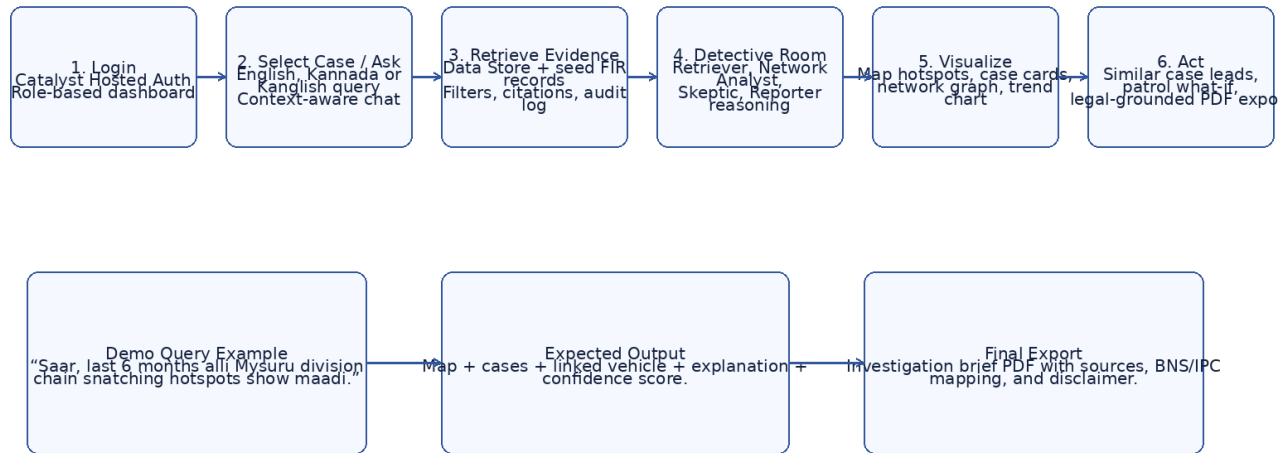
6. Full Feature Blueprint

Module	Purpose	Prototype Implementation
Role-Based Login	Investigator, Analyst, Supervisor, Admin roles with different module access.	Use Catalyst Hosted Authentication or demo JWT fallback.
Conversational Crime Search	Ask questions about FIRs, crime type, district, time window, accused links, and status.	Text query first; Kannada/Kanglish demo phrases mapped through prompt examples.

Detective Room Agents	Visible reasoning cards: Retriever, Network Analyst, Skeptic, Reporter.	Can be implemented as modular functions even if only one LLM is used.
KAVACH Crime DNA Engine	Find similar cases using MO, time, location, crime type, accused links, and FIR narrative similarity.	Start with weighted scoring; vector embeddings optional.
Criminal Network Graph	Visualize accused-case-location-phone-vehicle-account relationships.	Use React Flow/Cytoscape/D3; network data from relations table.
Hotspot Map	Show crime clusters on district/station map.	Use Leaflet/Folium-style coordinates from seed data.
What-If Patrol Simulation	Estimate hotspot coverage improvement with extra patrol units.	Rule-based prototype simulation with clear disclaimer.
Legal-Grounded PDF Report	Export chat answer, evidence sources, BNS/IPC mapping, confidence, audit timestamp.	Use SmartBrowz or HTML-to-PDF fallback.
Audit Log	Store every query, user role, timestamp, data filters, and generated report ID.	Catalyst Data Store AuditLogs table.

7. End-to-End User Journey: Login to PDF Export

End-to-End Prototype Journey: Login to Report Export



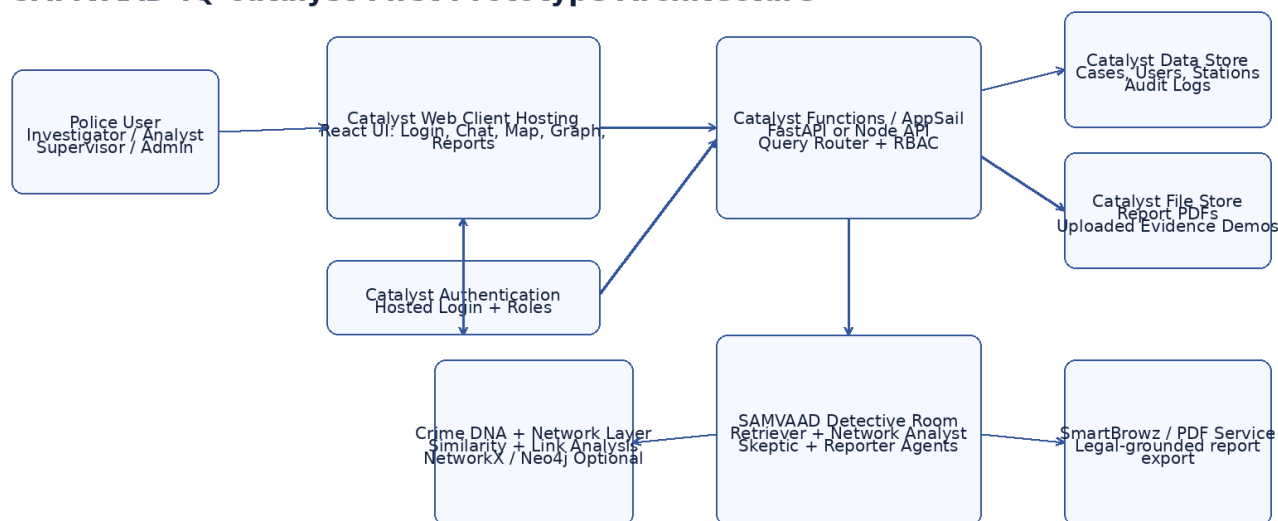
7.1 Complete workflow

1. User opens the Catalyst-hosted public prototype URL.
2. User logs in using Catalyst Hosted Authentication or the demo role selector credentials listed in the seed data.
3. The app lands on a role-aware dashboard. Investigator sees case search and investigation tools; Supervisor sees patrol what-if and reports; Analyst sees trends and maps.
4. User enters a natural-language query such as “Last 6 months alli Mysuru division chain snatching hotspots show maadi.”
5. The Query Router extracts intent: crime type = chain snatching, district = Mysuru, time window = last six months, output = hotspot map.

6. The Retriever Agent fetches matching records from Catalyst Data Store or local seed JSON fallback.
7. The Network Analyst checks shared accused IDs, vehicle hashes, phone hashes, and location proximity.
8. The Skeptic Agent flags weak evidence and adds a human-review disclaimer.
9. The Reporter Agent returns a clean answer with evidence cards, confidence score, and next-step suggestions.
10. The user opens Map, Graph, Similar Case, or What-If tabs from the same conversation.
11. User clicks Generate PDF. The report includes query, result, evidence trail, graph/map summary, BNS/IPC section mapping, confidence, and timestamp.
12. A report record is stored in File Store/Data Store and can be downloaded or shown in the demo video.

8. Catalyst-First System Architecture

SAMVAAD-IQ Catalyst-First Prototype Architecture



Deployment rule: public prototype link should be generated from Zoho Catalyst hosting/AppSail, then pasted into the hackathon

8.1 Catalyst component mapping

Catalyst Component	Use in SAMVAAD-IQ	Why
Web Client Hosting	Host the React frontend and generate public prototype link.	Matches submission requirement: deployed prototype link should be from Catalyst.
Catalyst Authentication	Login, sign-up/invite, and role-based access.	Provides official user and role management.
Data Store	Store cases, stations, accused, relations, audit logs, and reports metadata.	Relational data fits FIR-style records.
Functions or AppSail	Backend APIs, query router, similarity scoring, report generation.	Functions are fast for simple APIs; AppSail is better for FastAPI/Python services.
File Store	Store generated PDF reports and uploaded demo assets.	Keeps report files accessible to authorized users.
SmartBrowz PDF & Screenshot	Generate visual PDF reports from HTML templates.	Good for professional intelligence brief export.
API Gateway	Expose clean API endpoints and route web requests.	Separates frontend from backend resources.

Recommended build choice

For fastest hackathon delivery: use React on Catalyst Web Client Hosting + Node.js Catalyst Functions + Catalyst Data Store. If the team strongly prefers Python/FastAPI, deploy backend through Catalyst AppSail and keep frontend on Web Client Hosting.

9. Database Design and Manual Synthetic Data

The prototype must work reliably during testing without real police data. Therefore, SAMVAAD-IQ uses a manual synthetic dataset that contains planted patterns. The data is fictional, anonymized, and designed to demonstrate the platform's intelligence features.

9.1 Tables

Table	Important Fields	Purpose
Users	email, role, access_level, active_status	Demo roles and role permissions.
PoliceStations	station_id, station_name, district, lat, lon	Map and district-level filtering.
Cases	fir_id, district, station_id, crime_type, date, time, lat, lon, status, mo, summary, bns_sections	Main FIR-style case dataset.
Accused	accused_id, display_name, age_group, known_mo, prior_case_count, risk_note	Synthetic offender/person nodes.
Relations	source, target, type, weight	Graph edges for criminal network visualization.
AuditLogs	log_id, user_email, query, intent, timestamp, result_count, report_id	Explainability and governance.
Reports	report_id, user_email, title, generated_at, file_id, source_firs	PDF report metadata.

9.2 Demo login data

Email	Password	Role	Access
admin@samvaad.local	demo123	Admin	All modules, data seeding, user audit logs
investigator@samvaad.local	demo123	Investigator	Case search, similar cases, network graph, report generation
analyst@samvaad.local	demo123	Analyst	Dashboard, maps, trend analysis, model outputs
supervisor@samvaad.local	demo123	Supervisor	All insights, approval notes, patrol simulation, exports

9.3 Manual seed case examples

FIR ID	District	Crime Type	Date/Time	Status	Main Node	MO Summary
FIR-2025-BLR-001	Bengaluru South	Motorcycle Theft	2025-11-08 21:40	Under Investigation	A-112	Two-person team targets bikes parked near metro exit, breaks handle lock, uses duplicate ignition key.
FIR-2025-BLR-014	Bengaluru South	Motorcycle Theft	2025-11-16 22:15	Open	A-112	Two-person team targets commuter bikes

Synthetic data only. For hackathon prototype and evaluation demo.

						near bus stop, breaks handle lock, moves vehicle towards outer rin...
FIR-2025-MYS-006	Mysuru	Chain Snatching	2025-10-03 19:25	Charge Sheet Filed	A-301	Rider-pillion pair on scooter approaches pedestrian from left side and snatches gold chain near market ro...
FIR-2025-MYS-011	Mysuru	Chain Snatching	2025-10-10 19:45	Under Investigation	A-301	Rider-pillion pair on scooter approaches pedestrian from left side; target selected near dimly-lit lane.
FIR-2025-HUB-009	Hubballi-Dharwad	UPI Fraud	2025-09-19 14:05	Open	A-450	Caller impersonates bank support, sends malicious UPI collect request, transfers amount through mule acco...
FIR-2025-MNG-004	Mangaluru	UPI Fraud	2025-09-22 13:55	Open	A-450	Caller impersonates bank support, uses same wording and mule account pattern as earlier fraud case.
FIR-2025-BLR-027	Bengaluru South	House Burglary	2025-08-22 02:30	Open	A-777	Entry through rear window between 1 AM and 4 AM; targets locked independent houses.
FIR-2025-BLR-033	Bengaluru South	House Burglary	2025-08-29 02:10	Open	A-777	Entry through rear window between 1 AM and 4 AM; CCTV shows same small goods vehicle nearby.

9.4 Planted patterns for demo discovery

Pattern	Synthetic Evidence	Demo Query
Motorcycle theft cluster	FIR-2025-BLR-001 and FIR-2025-BLR-014 share A-112, PH-HASH-711, evening transit-location MO.	Are FIR-2025-BLR-001 and FIR-2025-BLR-014 connected?
Mysuru chain-snatching pattern	FIR-2025-MYS-006 and FIR-2025-MYS-011 share A-301, scooter vehicle KA-09-SYN-3001, 7 PM pattern.	Show Mysuru chain-snatching hotspots.
UPI fraud network	Hubballi and Mangaluru cases share A-450, phone hash, bank impersonation MO, and mule account pattern.	Show financial fraud links across districts.
Cold/similar burglary pattern	Two Bengaluru South burglary cases	Find similar cases to FIR-2025-BLR-027.

	share late-night timing, rear-window entry, and same vehicle hash.	
--	--	--

10. AI/ML Logic and Detective Room Agents

10.1 Detective Room module design

Agent	Responsibility	Prototype Logic
Retriever Agent	Fetch relevant FIR rows and metadata.	Keyword filters + intent parsing + Data Store query.
Network Analyst Agent	Find shared accused, phones, vehicles, accounts, locations.	Graph traversal over Relations table.
Crime DNA Agent	Score similarity between two or more cases.	Weighted score over crime type, MO tokens, time window, geography, repeated entities.
Skeptic Agent	Prevent overclaiming and hallucination.	Adds confidence limits, “not proof” disclaimer, missing-data notes.
Reporter Agent	Create final human-readable answer.	Formats answer, evidence cards, next steps, and report-ready text.

10.2 Crime DNA scoring formula

The hackathon version can use a transparent weighted score before adding embeddings. This is easier to debug and explain to judges.

```
crime_dna_score =
  0.25 * crime_type_match +
  0.20 * modus_operandi_similarity +
  0.15 * time_window_similarity +
  0.15 * location_proximity_score +
  0.15 * shared_entity_score +
  0.10 * status_or_investigation_pattern
```

Interpretation:

0.80 to 1.00 = strong investigative lead

0.60 to 0.79 = moderate similarity, needs verification

0.40 to 0.59 = weak similarity

below 0.40 = likely unrelated

10.3 Intent categories

Intent	Example Query	Response Type
HOTSPOT_QUERY	Show Mysuru chain-snatching hotspots.	Map + summary + top stations.
CASE_LINK_QUERY	Are these two FIRs connected?	Network graph + shared evidence.
SIMILAR_CASE_QUERY	Find similar cases to FIR-2025-BLR-027.	Crime DNA score and similar FIR list.
PATROL_WHAT_IF	What if 3 patrols are added at night?	Before/after coverage estimate.
LEGAL_REPORT	Generate a report for this cluster.	PDF-ready intelligence brief.
STATUS_QUERY	How many cases are open in Bengaluru South?	Count, filters, and dashboard cards.

11. API Design and Prototype Response Engine

11.1 REST endpoints

Method	Endpoint	Purpose	Input
POST	/api/auth/role-session	Create demo session or validate Catalyst auth token.	email, selectedRole
GET	/api/dashboard/summary	Get dashboard cards and charts.	role, district, dateRange
POST	/api/query	Main conversational query router.	query, userRole, conversationId
GET	/api/cases/:firId	Get FIR details.	firId
GET	/api/cases/:firId/similar	Get Crime DNA matches.	firId
GET	/api/network/:entityId	Get graph nodes and edges.	entityId
POST	/api/patrol/simulate	Run what-if patrol simulation.	district, timeWindow, patrolUnits
POST	/api/reports/generate	Generate PDF investigation brief.	conversationId, firIds, graphSnapshot
GET	/api/reports/:reportId	Download/view report.	reportId

11.2 Query response JSON shape

```
{
  "conversationId": "CONV-001",
  "intent": "CASE_LINK_QUERY",
  "answer": "FIR-2025-BLR-001 and FIR-2025-BLR-014 show a strong investigative link...",
  "confidence": 0.88,
  "evidence": [
    {"type": "shared_accused", "value": "A-112"},
    {"type": "shared_phone_hash", "value": "PH-HASH-711"},
    {"type": "similar_mo", "value": "motorcycle theft near transit points after 9 PM"}
  ],
  "visuals": {
    "mapPoints": [],
    "networkNodes": [],
    "networkEdges": [],
    "charts": []
  },
  "disclaimer": "Investigative lead only. Requires human verification."
}
```

12. Frontend Pages and UI Screens

Page / Screen	Components	Behavior
Login	Logo, role selector, Catalyst login button, demo credential cards.	Authenticates user and redirects based on role.
Dashboard	Crime cards, district filter, trend chart, hotspot list.	Loads summary from /api/dashboard/summary.
Investigation Chat	Chat input, voice button, Detective Room panel, evidence cards.	Main user workspace for queries.
Map View	Leaflet map, filters, hotspot markers, station cards.	Receives mapPoints from query response.
Network View	React Flow/Cytoscape graph, entity sidebar, edge evidence.	Shows accused-case-location-phone-account connections.
Similar Cases	Selected FIR card, Crime DNA score list, factor explanation.	Finds behaviorally similar FIR records.

Patrol What-If	District selector, time window, patrol units slider, before/after cards.	Runs rule-based coverage simulation.
Reports	Report preview, generate PDF button, download history.	Shows report status and download link.
Admin Seed Data	Import seed JSON, table viewer, reset demo data.	Admin-only helper for evaluation setup.

13. Zoho Catalyst Deployment Plan

The submission form requires the prototype deployment to be done using Catalyst platform by Zoho. Therefore, the final public prototype URL should come from Catalyst Web Client Hosting or Catalyst AppSail, not Vercel, Render, Netlify, or any other hosting platform.

13.1 Recommended deployment architecture

- Frontend: React + Vite deployed through Catalyst Web Client Hosting.
- Backend option A: Node.js Catalyst Functions for /api routes. This is fastest for Catalyst-native deployment.
- Backend option B: Python FastAPI deployed through Catalyst AppSail if the team needs Python libraries for similarity scoring or ML.
- Database: Catalyst Data Store tables for cases, stations, users, relations, audit logs, and reports.
- Authentication: Catalyst Hosted Authentication with roles, or demo role selection for fast prototype if auth setup becomes a bottleneck.
- PDF: SmartBrowz if enabled; otherwise generate HTML report and allow browser print-to-PDF as a fallback while keeping report data stored in Catalyst.

13.2 Catalyst setup checklist

13. Create a Zoho Catalyst account and project named samvaad-iq.
14. Install Catalyst CLI locally and log in.
15. Initialize project resources: client, functions, and/or AppSail service.
16. Create Data Store tables: Cases, PoliceStations, Accused, Relations, AuditLogs, Reports.
17. Configure Authentication roles: Admin, Investigator, Analyst, Supervisor.
18. Add seed data manually through console or through a seed API function.
19. Build frontend locally and test with seed JSON.
20. Deploy client and backend resources using Catalyst CLI or console.
21. Open deployed Catalyst public link and verify all demo queries.
22. Paste Catalyst deployed link into the submission portal.

13.3 Suggested CLI command flow

```
# Install Catalyst CLI globally if not already installed
npm install -g zcatalyst-cli

# Login and initialize project
catalyst login
mkdir samvaad-iq && cd samvaad-iq
catalyst init

# Select/create Catalyst project, then initialize client and functions
catalyst client:setup
catalyst functions:setup
```

```
# Local test
npm run dev
catalyst serve

# Build frontend
cd client
npm install
npm run build
cd ..

# Deploy all resources to Catalyst
catalyst deploy

# After deployment, copy the generated Web Client Hosting/AppSail URL
# and submit it as the Prototype Deployed Link.
```

13.4 Catalyst Data Store table creation

Create columns as text/number/date fields in Catalyst console. For quick demo, string fields can store JSON-like arrays for `accused_ids` and `tags`. If using backend functions, normalize into relations table for graph queries.

Table	Columns to Create
Cases	<code>fir_id</code> , <code>district</code> , <code>station_id</code> , <code>crime_type</code> , <code>date</code> , <code>time</code> , <code>lat</code> , <code>lon</code> , <code>status</code> , <code>mo</code> , <code>case_summary</code> , <code>bns_sections</code> , <code>accused_ids</code> , <code>victim_id</code> , <code>vehicle</code> , <code>phone_hash</code>
PoliceStations	<code>station_id</code> , <code>station_name</code> , <code>district</code> , <code>lat</code> , <code>lon</code>
Accused	<code>accused_id</code> , <code>display_name</code> , <code>age_group</code> , <code>known_mo</code> , <code>prior_case_count</code> , <code>risk_note</code>
Relations	<code>source</code> , <code>target</code> , <code>type</code> , <code>weight</code>
AuditLogs	<code>user_email</code> , <code>role</code> , <code>query</code> , <code>intent</code> , <code>timestamp</code> , <code>result_count</code> , <code>confidence</code>
Reports	<code>report_id</code> , <code>title</code> , <code>user_email</code> , <code>generated_at</code> , <code>source_firs</code> , <code>file_id</code> , <code>status</code>

14. GitHub Public Repository Plan

```
samvaad-iq/
├── README.md
├── catalyst.json
├── client/
│   ├── package.json
│   └── src/
│       ├── pages/Login.jsx
│       ├── pages/Dashboard.jsx
│       ├── pages/InvestigationChat.jsx
│       ├── pages/MapView.jsx
│       ├── pages/NetworkView.jsx
│       ├── pages/Reports.jsx
│       ├── components/DetectiveRoom.jsx
│       ├── components/CrimeDNA.jsx
│       └── data/demoSeed.json
├── functions/
│   ├── query_router/
│   ├── dashboard_api/
│   └── report_generator/
```

Synthetic data only. For hackathon prototype and evaluation demo.

```

├── seed_data/
├── appsail/          # Optional if using Python/FastAPI
│   ├── main.py
│   ├── requirements.txt
│   └── app-config.json
├── seed/
│   ├── SAMVAAD-IQ_Prototype_Seed_Data.json
│   └── import_instructions.md
├── docs/
│   ├── architecture.png
│   ├── demo_script.md
│   └── prototype_deck.pdf
└── screenshots/
    ├── login.png
    ├── dashboard.png
    ├── detective_room.png
    ├── network_graph.png
    └── report_export.png

```

- README must include: project overview, selected challenge, deployed Catalyst link, demo video link, setup steps, seed credentials, tech stack, and screenshots.
- Never commit real secrets. Use .env.example for API keys and Catalyst project details.
- Keep repository public before submitting the GitHub link.
- Add license and disclaimer: synthetic data only; not for production policing without legal, ethical, and technical review.

15. Testing Matrix with Expected Responses

Role	Test Query	Expected Answer	Expected Visual
Investigator	Last 6 months alli Mysuru division chain snatching hotspots show maadi	Two synthetic chain-snatching cases appear around Lashkar and VV Puram between 19:00 and 20:00. The repeated scooter vehicle KA-09-SYN-3001 and shared MO indicate a possible linked pattern. Confidence: 0.86.	Hotspot map with two Mysuru points and a case card list.
Investigator	Are FIR-2025-BLR-001 and FIR-2025-BLR-014 connected?	Yes. Both are Bengaluru South motorcycle thefts after 9 PM, share accused A-112, phone hash PH-HASH-711, similar transit-location MO, and close time pattern. Treat this as an investigative lead, not final proof.	Network graph: A-112 -> two FIR nodes -> shared phone hash -> two police stations.
Analyst	Find similar cases to FIR-2025-BLR-027	FIR-2025-BLR-033 is the closest match. Similarity drivers: late-night timing, rear-window entry, independent-house target, same vehicle hash KA-03-SYN-7777, and Bengaluru South geography. Similarity score: 0.84.	Similar-case panel with Crime DNA factors.
Supervisor	What if 3 patrol units are added to Jayanagar and Banashankari from 9 PM to 12 AM?	Prototype simulation estimates higher coverage of motorcycle-theft hotspots from 56% to 82%. Recommended patrol focus: metro exit, bus stand, and outer ring road exit corridors. This is a resource-planning estimate, not a guarantee.	Before/after patrol coverage cards and route suggestion.
Supervisor	Generate PDF report for the	Report generated with case list,	PDF export button and report

Synthetic data only. For hackathon prototype and evaluation demo.

	motorcycle theft cluster	evidence links, graph image, BNS mapping, confidence notes, human-review disclaimer, and audit timestamp.	preview.
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16. Demo Video Storyboard

Time	Scene	What to Show	Voiceover
0:00-0:20	Problem opening	Excel sheets / files / map blur, then SAMVAAD-IQ logo.	Crime data exists, but patterns are lost across FIRs, stations, and manual reports.
0:20-0:40	Login	Catalyst-hosted URL, login as Investigator.	SAMVAAD-IQ is deployed on Zoho Catalyst with role-based access.
0:40-1:10	Dashboard + query	Ask Kanglish query about Mysuru chain snatching.	The officer can ask naturally in English, Kannada, or Kanglish.
1:10-1:45	Detective Room	Agents show retrieval, network analysis, skeptic notes, final answer.	The AI explains how it reached the answer and shows the source cases.
1:45-2:15	Network graph	Show linked accused, FIRs, scooter vehicle, station nodes.	Hidden relationships become visible instantly.
2:15-2:40	Crime DNA	Find similar case to burglary FIR.	Crime DNA compares modus operandi, time, location, and shared entities.
2:40-3:05	Patrol what-if	Slider: 3 patrol units, before/after coverage cards.	Supervisors can test patrol strategies before field deployment.
3:05-3:30	PDF report + end	Generate report, show download, submission-ready screen.	Every insight is evidence-backed, auditable, and exportable.

17. Prototype Deck Structure

Slide	Title	Content
1	SAMVAAD-IQ	Tagline, team name, selected challenge.
2	Problem	Fragmented crime records, slow analysis, reactive policing.
3	Why PS2	Conversational AI subsumes analytics and visualization.
4	Solution Overview	Conversational Crime Intelligence OS.
5	System Architecture	Catalyst-first architecture diagram.
6	Core Innovation 1	Detective Room multi-agent reasoning.
7	Core Innovation 2	KAVACH Crime DNA similar case engine.
8	Core Innovation 3	Network graph + hotspot map + patrol what-if.
9	Prototype Demo Flow	Login to query to map/graph to PDF.
10	Zoho Catalyst Deployment	Web client, functions/AppSail, Data Store, Auth, report export.
11	Security and Ethics	Synthetic data, human-in-loop, no automated guilt prediction.
12	Impact and Roadmap	Operational value, future multilingual/real-data integration.

18. Security, Privacy, Ethics, and Legal Safeguards

- Use synthetic data only for prototype and demo. Do not upload real FIRs or PII unless official permission and governance exist.
- Do not claim that the AI predicts guilt or future criminal behavior. It only generates investigative hypotheses and prioritization leads.
- Every answer must include evidence references, confidence score, data limitations, and a human-review disclaimer.
- Role-based access prevents unauthorized users from viewing sensitive case-level details.
- Audit logs store user, query, time, filters, results, and report exports.
- Bias control: avoid demographic-based suspect prediction; use case-level MO and network evidence, not protected personal attributes.
- Legal-grounded reports should map sections as informational support only. Final legal interpretation remains with authorized officers/legal experts.

Important demo language

Say: “SAMVAAD-IQ finds investigative leads from patterns in records.” Avoid saying: “SAMVAAD-IQ proves someone is guilty” or “predicts a person will commit crime.”

19. Implementation Timeline and Scope Control

Phase	Duration	Build Items	Must Work Before Moving On
Day 1	6-8 hrs	Repo setup, React layout, Catalyst project, seed JSON, login screen.	Login and dashboard render with static data.
Day 2	8-10 hrs	Data Store import, query router, dashboard API, chat response logic.	Queries return correct evidence from manual data.
Day 3	8-10 hrs	Map view, network graph, Crime DNA similar case scoring.	All 5 test queries produce expected responses.
Day 4	6-8 hrs	Patrol what-if, PDF/HTML report, audit logs.	Report export works and audit log is stored.
Day 5	4-6 hrs	Catalyst deployment, GitHub polish, demo video, deck PDF.	Public links work in incognito browser.

19.1 Scope traps to avoid

- Do not build full real Kannada NLP if time is low. Map 10-15 Kannada/Kanglish demo phrases to intents first.
- Do not build real ML forecasting before the core demo works. Use transparent rule-based risk and label it as prototype simulation.
- Do not integrate real financial APIs. Use synthetic account hashes and graph relations.
- Do not overbuild authentication if Catalyst Hosted Auth setup blocks progress. Use demo role cards as fallback but keep Catalyst deployment mandatory.
- Do not chase 3D visualizations. A clean 2D network graph and map will impress more if reliable.

20. Roadmap and Future Extensions

Stage	Future Capability
-------	-------------------

Synthetic data only. For hackathon prototype and evaluation demo.

Prototype	Synthetic FIR data, role login, chat, map, graph, Crime DNA, report export.
Pilot	Connect sanitized station-level historical data, stronger Kannada NLP, better audit workflow.
Advanced	Graph database at scale, embedding-based semantic search, alert subscriptions, supervised model validation.
Production	Formal data governance, bias audit, legal approval, secure hosting, incident response, integration with SCRB workflows.

Appendix A. Seed Data Summary

A full machine-readable seed data file is provided separately as SAMVAAD-IQ_Prototype_Seed_Data.json. The JSON contains demo users, police stations, FIR-style cases, accused nodes, and relations. Import it manually into Catalyst Data Store or load it as a local fallback during prototype testing.

```
{
  "users": [
    {
      "email": "admin@samvaad.local",
      "password": "demo123",
      "role": "Admin",
      "access": "All modules, data seeding, user audit logs"
    },
    {
      "email": "investigator@samvaad.local",
      "password": "demo123",
      "role": "Investigator",
      "access": "Case search, similar cases, network graph, report generation"
    }
  ],
  "police_stations": [
    {
      "station_id": "PS-BLR-JYN",
      "station_name": "Jayanagar Police Station",
      "district": "Bengaluru South",
      "lat": 12.925,
      "lon": 77.5938
    },
    {
      "station_id": "PS-BLR-BSK",
      "station_name": "Banashankari Police Station",
      "district": "Bengaluru South",
      "lat": 12.9182,
      "lon": 77.5736
    }
  ],
  "cases": [
    {
      "fir_id": "FIR-2025-BLR-001",
      "district": "Bengaluru South",
      "station_id": "PS-BLR-JYN",
      "crime_type": "Motorcycle Theft",
      "date": "2025-11-08",

```



```

"time": "21:40",
"lat": 12.9253,
"lon": 77.594,
"status": "Under Investigation",
"mo": "Two-person team targets bikes parked near metro exit, breaks handle lock, uses duplicate ignition key.",
"case_summary": "Synthetic report: black motorcycle removed from public parking near Jayanagar metro after 9 PM.",
"bns_sections": "303(2), 317",
"accused_ids": [
  "A-112",
  "A-205"
],
"victim_id": "V-501",
"vehicle": "KA-05-SYN-1122",
"phone_hash": "PH-HASH-711"
},
{
  "fir_id": "FIR-2025-BLR-014",
  "district": "Bengaluru South",
  "station_id": "PS-BLR-BSK",
  "crime_type": "Motorcycle Theft",
  "date": "2025-11-16",
  "time": "22:15",
  "lat": 12.9186,
  "lon": 77.573,
  "status": "Open",
  "mo": "Two-person team targets commuter bikes near bus stop, breaks handle lock, moves vehicle towards outer ring road.",
  "case_summary": "Synthetic report: motorcycle stolen near Banashankari bus stand during evening crowd.",
  "bns_sections": "303(2), 317",
  "accused_ids": [
    "A-112"
  ],
  "victim_id": "V-518",
  "vehicle": "KA-05-SYN-1189",
  "phone_hash": "PH-HASH-711"
}
],
"relations": [
  {
    "source": "A-112",
    "target": "FIR-2025-BLR-001",
    "type": "ACCUSED_IN",
    "weight": 0.95
  },
  {
    "source": "A-112",
    "target": "FIR-2025-BLR-014",
    "type": "ACCUSED_IN",
    "weight": 0.92
  }
]
}

```

Appendix B. Prompt Templates

B.1 System prompt for Reporter Agent

You are SAMVAAD-IQ Reporter Agent. Answer only from the provided case records and evidence. Never invent facts. Separate confirmed data from investigative hypotheses. Include confidence, source FIR IDs, reasoning factors, and a human-review disclaimer. Avoid saying any person is guilty. Use simple professional language suitable for police users.

B.2 Skeptic Agent prompt

Review the draft answer. Identify weak links, missing evidence, possible bias, and overclaiming. Add one concise warning if the answer implies guilt, future crime, or certainty beyond data. Output: risk flags, corrected wording, and confidence adjustment.

B.3 Crime DNA Agent prompt

Compare the selected FIR against candidate FIRs using crime type, modus operandi, time window, location proximity, shared accused/vehicle/phone/account, and narrative similarity. Return a score from 0 to 1 and the top evidence factors. This is for investigative lead generation only.

Appendix C. Submission Checklist

Item	Status Check
Challenge selected	Problem Statement 2 selected in portal.
Prototype Brief	Paste Section 4 text; verify under 1024 characters.
GitHub Public Repo	Repo public, README complete, no secrets, screenshots added.
Catalyst Deployed Link	Open in incognito; login and demo queries work.
Demo Video	Public Google Drive or unlisted YouTube link; no access restriction.
Prototype Deck PDF	Export deck as PDF and upload.
Manual Data	Seed JSON committed and/or imported into Catalyst Data Store.
Final Demo Query Test	Run all test queries in Section 15 before submission.

References and Official Deployment Notes

- Zoho Catalyst Web Client Hosting documentation: <https://docs.catalyst.zoho.com/en/cloud-scale/help/web-client-hosting/introduction/>
- Zoho Catalyst CLI deploy resources documentation: <https://docs.catalyst.zoho.com/en/cli/v1/deploy-resources/introduction/>
- Zoho Catalyst AppSail documentation: <https://docs.catalyst.zoho.com/en/serverless/help/appsail/introduction/>
- Zoho Catalyst Authentication documentation: <https://docs.catalyst.zoho.com/en/cloud-scale/help/authentication/introduction/>
- Zoho Catalyst Data Store documentation: <https://docs.catalyst.zoho.com/en/cloud-scale/help/data-store/introduction/>
- Zoho Catalyst SmartBrowz PDF and Screenshot documentation: <https://docs.catalyst.zoho.com/en/smartbrowz/help/pdfnscreenshot/introduction/>